HEXLEY CHEMICALS

Past Permit Decisions Casebook

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1. About this casebook

Eight past reviews, kept as worked examples of the Control of Work Policy, the Permit Rules and the permit standards in practice. Each case records the request, its answers as they were written, each requirement's result, the SIMOPS triggers, the decisions open, the decision taken and why, and how the work went.

The casebook is background reading for Permit Coordinators and Area Authorities. It is never cited to a performing authority as a rule, and a request is always reviewed under the latest issue of its own permit type's standard, whatever an earlier case decided.

2. A hot work permit read against the confined space standard

Request PTW0050214, sent from the site portal, asked for a permit for a weld repair on a solvent line in Reactor House A while the reactors were shut down. It was reviewed under the hot work standard.

The request gave: permit type "Hot work"; linked permit type "No linked permit"; plant area "Reactor House A"; classification "Zone 1"; unit status "Shut down"; location "Indoors"; risk assessment "Yes"; flammable gas "8"; oxygen "20.9"; clearance "12"; fire watch "60".

The coordinator who opened the review first read the confined space entry standard, whose sentences are almost the same, and found the flammable gas reading within it. Read against the hot work standard, the request's own, 8 percent of the lower explosive limit is above the figure, so the flammable gas requirement was not met. The live permit register showed an electrical isolation live in the reactor house, which conflicts with nothing, and the unit was shut down.

- **Risk assessment**, hot work standard - Met
- **Flammable gas**, hot work standard - Not met
- **Oxygen**, hot work standard - Met
- **Combustibles cleared**, hot work standard - Met
- **Fire watch**, hot work standard - Met

The assessment was RETURN TO REQUESTER. No SIMOPS trigger applied. The decisions open were RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority returned it to its requester. The line was purged with nitrogen and tested again at 0 percent, the request was sent again, and the permit was issued.

3. Readings in milligrams per cubic metre

Request PTW0050377, sent from scanned paper as 50377, asked for a permit for cleaning out a sump in the Effluent Treatment plant. It was reviewed under the confined space entry standard.

The request gave: permit type "CSE"; linked permit type "NIL"; plant area "Effluent Treatment"; classification "ZONE-2"; unit status "S/D"; location "OUT"; risk assessment "Y"; flammable gas "2 % LEL"; oxygen "20.9 % O2"; hydrogen sulphide "6.8 mg/m3"; carbon monoxide "21.1 mg/m3"; rescue plan "Y".

Scanned paper writes the toxic readings in milligrams per cubic metre: 6.8 mg/m3 of hydrogen sulphide is 4.8 ppm and 21.1 mg/m3 of carbon monoxide is 18.0 ppm, both within the confined space entry standard's figures. Read as ppm, the same figures would have failed both requirements and sent a sound request back. Nothing was live in the area, and the plant was shut down.

- **Risk assessment**, confined space entry standard - Met
- **Flammable gas**, confined space entry standard - Met
- **Oxygen**, confined space entry standard - Met
- **Hydrogen sulphide**, confined space entry standard - Met
- **Carbon monoxide**, confined space entry standard - Met
- **Rescue plan**, confined space entry standard - Met

The assessment was ISSUE PERMIT. No SIMOPS trigger applied. The decisions open were ISSUE PERMIT, RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority issued the permit, and the permit document gave both readings in ppm. The clean-out was completed without a stop.

4. A conflicting permit live in the area

Request PTW0050452, sent from the contractor app as PTW-50452, asked for a permit for an internal inspection of a column in the Distillation Unit during a shutdown. It was reviewed under the confined space entry standard.

The request gave: permit type "Confined Space Permit"; linked permit type "No link"; plant area "Distillation Unit"; classification "Z2"; unit status "Off-line"; location "Outside"; risk assessment "true"; flammable gas "0%LEL"; oxygen "20.9%"; hydrogen sulphide "0 ppm"; carbon monoxide "0 ppm"; rescue plan "true".

Every requirement was met. But the live permit register showed two permits live in the Distillation Unit: a work at height permit, which conflicts with nothing, and a hot work permit, which conflicts with confined space entry. Sparks from the welding could reach an open manway, and an entrant cannot get out fast, so the conflicting live permit trigger applied.

- **Risk assessment**, confined space entry standard - Met
- **Flammable gas**, confined space entry standard - Met
- **Oxygen**, confined space entry standard - Met
- **Hydrogen sulphide**, confined space entry standard - Met
- **Carbon monoxide**, confined space entry standard - Met
- **Rescue plan**, confined space entry standard - Met

The assessment was REFER FOR SIMOPS REVIEW. The SIMOPS triggers that applied: conflicting live permit. The decisions open were RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority referred it for a SIMOPS review. The panel agreed that the welding would stop, and its area be cleared, while anyone was inside the column, and the permit was issued with that control.

5. A suspended permit does not count

Request PTW0050518, sent from the site portal, asked for a permit for a trial hole beside the Weighbridge. It was reviewed under the excavation standard.

The request gave: permit type "Excavation"; linked permit type "No linked permit"; plant area "Weighbridge"; classification "Non-hazardous"; unit status "No process plant"; location "Outdoors"; risk assessment "Yes"; depth "1.1"; services located "Yes"; barriers "Yes".

The live permit register showed a hot work permit at the Weighbridge, and hot work conflicts with excavation. But that permit had been suspended after a site alarm, its work stopped until the Area Authority revalidated it, so it did not count, and no SIMOPS trigger applied. At 1.1 m deep, the dig was not deeper than the excavation standard's figure, so its support was not asked for, and every requirement was met.

- **Risk assessment**, excavation standard - Met
- **Services located**, excavation standard - Met
- **Trench support**, excavation standard - Met
- **Barriers**, excavation standard - Met

The assessment was ISSUE PERMIT. No SIMOPS trigger applied. The decisions open were ISSUE PERMIT, RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority issued the permit, and noted that the hot work could not be revalidated while the trial hole was open without a review of its own.

6. A linked permit must meet both standards

Request PTW0050663, sent from the site portal, asked for a permit for pipework repairs at height on the Cooling Towers, with welding. It was reviewed under the work at height standard and, for its linked permit type, the hot work standard.

The request gave: permit type "Work at height"; linked permit type "Hot work"; plant area "Cooling Towers"; classification "Non-hazardous"; unit status "Shut down"; location "Outdoors"; risk assessment "Yes"; working height "7.5"; access "Mobile elevating work platform"; fall protection "Yes"; barriers "Yes"; wind "4.5"; flammable gas "0"; oxygen "20.9"; clearance "11"; fire watch "30".

Every requirement of the work at height standard was met, the platform reaching the work at 7.5 m in a light wind. The welding made hot work its linked permit type, and a request must meet its linked type's standard as well as its own: a fire watch of 30 minutes is short of the hot work standard's figure, so the fire watch was not met.

- **Risk assessment**, work at height standard - Met
- **Fall protection**, work at height standard - Met
- **Scaffold inspection**, work at height standard - Met
- **Barriers**, work at height standard - Met
- **Wind speed**, work at height standard - Met
- **Risk assessment**, hot work standard - Met
- **Flammable gas**, hot work standard - Met
- **Oxygen**, hot work standard - Met
- **Combustibles cleared**, hot work standard - Met
- **Fire watch**, hot work standard - Not met

The assessment was RETURN TO REQUESTER. No SIMOPS trigger applied. The decisions open were RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority returned it to its requester, who arranged a fire watch for the full time and sent it again, and the permit was issued with both standards met.

7. Hot work on a running unit through a linked permit

Request PTW0050729, sent from the contractor app as PTW-50729, asked for a permit for repairs to a steam line at height in Caustic Evaporation, with welding. It was reviewed under the work at height standard and, for its linked permit type, the hot work standard.

The request gave: permit type "Work at Height Permit"; linked permit type "Hot Work Permit"; plant area "Caustic Evaporation"; classification "Safe area"; unit status "Running"; location "Outside"; risk assessment "true"; working height "9 m"; access "Scaffold"; scaffold inspected "true"; fall protection "true"; barriers "true"; wind "5.5 m/s"; flammable gas "0%LEL"; oxygen "20.9%"; clearance "12 m"; fire watch "60 min".

Both standards were met, the scaffold inspected and tagged. Work at height on its own is never a live process unit, but the welding made hot work its linked permit type, and hot work on a process unit that is running is: the live process unit trigger applied. The electrical isolation live in the area conflicts with nothing.

- **Risk assessment**, work at height standard - Met
- **Fall protection**, work at height standard - Met
- **Scaffold inspection**, work at height standard - Met
- **Barriers**, work at height standard - Met
- **Wind speed**, work at height standard - Met
- **Risk assessment**, hot work standard - Met
- **Flammable gas**, hot work standard - Met
- **Oxygen**, hot work standard - Met
- **Combustibles cleared**, hot work standard - Met
- **Fire watch**, hot work standard - Met

The assessment was REFER FOR SIMOPS REVIEW. The SIMOPS triggers that applied: live process unit. The decisions open were RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority referred it for a SIMOPS review. The panel agreed with the control room which lines would be taken out of service while the welding went on, and the permit was issued with that control.

8. A voltage written in kilovolts

Request PTW0050841, sent from scanned paper as 50841, asked for a permit for maintenance of a motor starter on a high voltage board in the Compressor House. It was reviewed under the electrical isolation standard.

The request gave: permit type "EI"; linked permit type "NIL"; plant area "Compressor House"; classification "ZONE-2"; unit status "OPER"; location "IN"; risk assessment "Y"; system voltage "3.3 kV"; proved dead "Y"; personal locks "Y"; circuit earths "Y".

Scanned paper writes a voltage in kilovolts: 3.3 kV is 3,300 volts, above the electrical isolation standard's figure, so the circuit earths were asked for, and they were applied. Every requirement was met, but an isolation above 1,000 volts is a SIMOPS trigger on its own, because switching it affects the supply to the whole area. The running compressors made no difference: only hot work and confined space entry count as a live process unit.

- **Risk assessment**, electrical isolation standard - Met
- **Proved dead**, electrical isolation standard - Met
- **Personal locks**, electrical isolation standard - Met
- **Circuit earths**, electrical isolation standard - Met

The assessment was REFER FOR SIMOPS REVIEW. The SIMOPS triggers that applied: high voltage. The decisions open were RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority referred it for a SIMOPS review. The panel agreed the switching programme with the control room, and the permit was issued.

9. A gas test record that disagreed with the request

Request PTW0050906, sent from the site portal, asked for a permit for a weld repair on a pump baseplate in the Water Treatment plant. It was reviewed under the hot work standard.

The request gave: permit type "Hot work"; linked permit type "No linked permit"; plant area "Water Treatment"; classification "Non-hazardous"; unit status "Shut down"; location "Indoors"; risk assessment "Yes"; flammable gas "0"; oxygen "20.9"; clearance "15"; fire watch "90".

The gas test record said: "Flammable gas read 4% LEL at the work point."

Every requirement was met, and the work at height permit live beside it conflicts with nothing. But the gas test record said that flammable gas read 4 percent of the lower explosive limit where the request said 0, and the sentence was flagged DISAGREES WITH FORM.

- **Risk assessment**, hot work standard - Met
- **Flammable gas**, hot work standard - Met
- **Oxygen**, hot work standard - Met
- **Combustibles cleared**, hot work standard - Met
- **Fire watch**, hot work standard - Met

The assessment was ISSUE PERMIT. No SIMOPS trigger applied. The decisions open were ISSUE PERMIT, RETURN TO REQUESTER and REFER FOR SIMOPS REVIEW.

The Area Authority returned it to its requester to settle the reading, which RETURN TO REQUESTER, always open, allows. The performing authority found that the 4 percent had been read before a drum of cleaning solvent was taken out of the pump house, tested again at 0 percent, corrected the gas test record and sent the request again, and the permit was issued.